

A Geometric Analogue of Bertrand's Postulate over Imaginary Quadratic Rings

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August 6, 2026

Abstract

Bertrand's postulate asserts that for every $n > 1$, there exists a rational prime p with $n < p < 2n$. Fixing an integral basis $(1, \tau)$, we define a region $B_r(\alpha, (1+c)\alpha)$ using coordinate-wise inequalities and norm bounds. Under a thickness condition preventing the normalized basis coordinates of α from approaching either coordinate axis, we prove that $B_r(\alpha, (1+c)\alpha)$ contains a prime element for large enough $N(\alpha)$. The proof identifies an annular sector of uniform angular and radial width in the scaled region and applies an asymptotic angular prime distribution theorem for imaginary quadratic rings.

1 Introduction

Bertrand's postulate was conjectured by Bertrand in 1845 and first proved by Chebyshev using analytic methods ([**Cheb**]). Erdős later gave an elementary proof using estimates for binomial coefficients ([**Erdos**]).

A natural question is whether Bertrand-type results hold in other rings of integers. In $\mathbb{Z}$, the phrase between “ n and $2n$ ” is well defined. In $\mathbb{Z}[\sqrt{d}]$, however, the elements form a lattice in the complex plane, so there is no unique linear order. If one only requires

$$N(\alpha) < N(\pi) < N(2\alpha),$$

then π may lie in a completely different angular direction from α . Thus, a useful analogue of Bertrand's postulate in complex quadratic rings must include some geometric condition.

Klee, Loucks, Meek, Overcast, Stewart, and Tou studied this problem for the Gaussian integers in their paper *Bertrand's Postulate over the Gaussian Integers* ([**Klee**]).

Bertrand-type questions have also been studied for number fields more generally. Hulse and Murty proved results about the existence of prime ideals whose norms lie in intervals of the form

$$x \leq N(\mathfrak{p}) \leq B_K x$$

for suitable constants $B_K > 1$ depending on the field K ([**Hulse**]). Their result is a number-field analogue of Bertrand's postulate, but it is formulated in terms of norm intervals.

The analytic input used in this paper comes from angular distribution results for primes in imaginary quadratic rings. Classical work of Kubilius studied the distribution of Gaussian primes in sectors ([**Kubi**]). More directly, Sittinger states an angular prime-counting theorem for imaginary quadratic rings and derives the consequence that sufficiently large annular sectors contain prime elements ([**Sitt**]). This is the result used in the proof of the main theorem.

2 Preliminaries

Let K be an imaginary quadratic number field, and let $\mathcal{O}_K$ be its ring of integers. Choose an integral basis

$$\mathcal{O}_K = \mathbb{Z} + \mathbb{Z}\tau,$$

where $\tau \in \mathbb{C} \setminus \mathbb{R}$ and $\text{Im}(\tau) > 0$. Thus, every element of $\mathcal{O}_K$ can be written uniquely as

$$a_1 + a_2\tau, \quad a_1, a_2 \in \mathbb{Z}.$$

Geometrically, $\mathcal{O}_K$ forms a lattice in the complex plane.

Throughout this paper, for every complex number $z \in \mathbb{C}$, define $N(z) = |z|^2$.

A nonzero nonunit $\pi \in \mathcal{O}_K$ is a prime element if the principal ideal (π) is a nonzero prime ideal.

Definition 2.1. $\gamma = c_1 + c_2\tau$ lies between $\alpha = a_1 + a_2\tau$ and $\beta = b_1 + b_2\tau$, $a_1, a_2, b_1, b_2 \in \mathbb{R}$ if and only if

1. $\min\{N(\alpha), N(\beta)\} < N(\gamma) < \max\{N(\alpha), N(\beta)\}$;
2. $\min\{a_1, b_1\} < c_1 < \max\{a_1, b_1\}$;

$$3. \min\{a_2, b_2\} < c_2 < \max\{a_2, b_2\}.$$

We now introduce a thickness condition.

Definition 2.2. An element $\alpha = a_1 + a_2\tau$ is η -thick if and only if

1. $|a_1| \geq \eta R$;
2. $|a_2||\tau| \geq \eta R$,

where

$$R = \sqrt{N(\alpha)} = |\alpha|.$$

Define the continuous region

$$\mathcal{B}_\tau(\alpha, (1+c)\alpha) = \left\{ x + y\tau \left| \begin{array}{l} x, y \in \mathbb{R}, \\ x + \tau y \text{ lies between } \alpha \text{ and } (1+c)\alpha \end{array} \right. \right\}.$$

Let

$$B_\tau(\alpha, (1+c)\alpha) = \mathcal{B}_\tau(\alpha, (1+c)\alpha) \cap \mathcal{O}_K.$$

Theorem 2.1. Let $\mathcal{O}_K$ be an imaginary quadratic ring of integers. Fix $0 \leq \theta_1 < \theta_2 \leq 2\pi$, and let $\Pi(x; \theta_1, \theta_2)$ denote the number of prime elements $\rho \in \mathcal{O}_K$ satisfying

$$|\rho|^2 \leq x, \quad \theta_1 < \arg \rho < \theta_2.$$

[Sitt] gives us that

$$\Pi(x; \theta_1, \theta_2) \sim \frac{g(\theta_2 - \theta_1)}{2\pi h} \frac{x}{\log x},$$

where g is the number of units in $\mathcal{O}_K$, and h is the class number of $\mathcal{O}_K$.

We will use the following consequence of Proposition 3.1 of [Sitt]: for any $0 < a < b$ and any nonempty open angular interval $I = (\theta_1, \theta_2)$, there exists $X_0 > 0$ such that, for every $x \geq X_0$, the annular sector

$$\{z \in \mathbb{C} \mid ax < |z|^2 < bx, \arg z \in I\}$$

contains a prime element of $\mathcal{O}_K$.

Indeed,

$$\begin{aligned}\Pi(bx; I) - \Pi(ax; I) &\sim \frac{g|I|}{2\pi h} \left(\frac{bx}{\log(bx)} - \frac{ax}{\log(ax)} \right) \\ &\sim \frac{g|I|}{2\pi h} (b-a) \frac{x}{\log x} \longrightarrow \infty.\end{aligned}$$

Consequently, this difference is positive for all sufficiently large x .

3 Main Result

We begin with the geometric lemmas needed to construct a fixed annular sector inside the region.

Let

$$\alpha = a_1 + a_2\tau, \quad R = |\alpha|, \quad u = \frac{a_1}{R}, \quad v = \frac{a_2}{R}.$$

Then

$$|u + v\tau| = 1.$$

Since α is η -thick,

$$|u| \geq \eta, \quad |v||\tau| \geq \eta.$$

Lemma 3.1. For every η -thick α , the scaled region

$$W_{u,v} = \frac{1}{|\alpha|} \mathcal{B}_\tau(\alpha, (1+c)\alpha)$$

contains the ball $B(M_{u,v}, \delta)$, where

$$M_{u,v} = \left(1 + \frac{c}{2}\right) (u + v\tau)$$

and

$$\delta = \min \left\{ \frac{c}{4}, \frac{c\eta \sin \varphi}{4} \right\}.$$

Proof. Define

$$P_{u,v} = \left\{ X + Y\tau \left| \begin{array}{l} \min\{u, (1+c)u\} < X < \max\{u, (1+c)u\}, \\ \min\{v, (1+c)v\} < Y < \max\{v, (1+c)v\} \end{array} \right. \right\},$$

and

$$A_{1,1+c} = \{z \in \mathbb{C} \mid 1 < |z| < 1+c\}.$$

Scaling the coordinate inequalities and norm bounds in the definition of $\mathcal{B}_\tau(\alpha, (1+c)\alpha)$ by R gives

$$W_{u,v} = P_{u,v} \cap A_{1,1+c}.$$

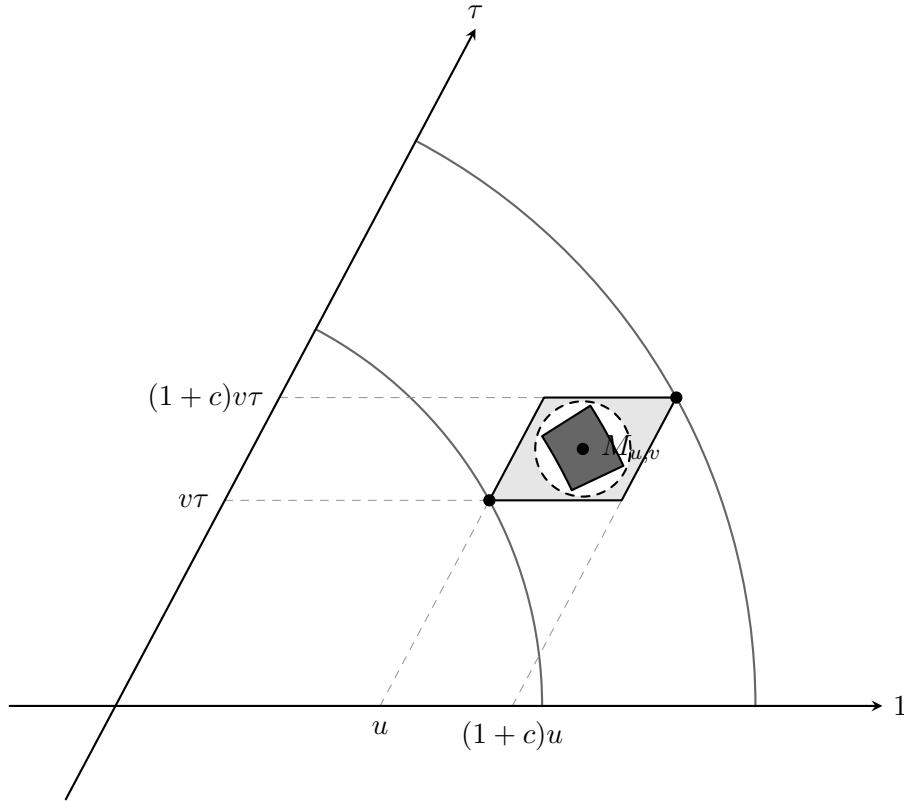


Figure 1: Annular region

We defined

$$M_{u,v} = \left(1 + \frac{c}{2}\right) (u + v\tau).$$

Because $|u + v\tau| = 1$,

$$|M_{u,v}| = 1 + \frac{c}{2}.$$

Hence

$$d(M_{u,v}, \partial A_{1,1+c}) = \frac{c}{2}.$$

Write $\varphi = \arg \tau$. The boundary of $P_{u,v}$ lies on the four lines

$$X = u, \quad X = (1 + c)u, \quad Y = v, \quad Y = (1 + c)v.$$

A line of the form $X = C$ is parallel to the direction τ . Therefore, the Euclidean distance from a point $X_0 + Y_0\tau$ to the line $X = C$ is

$$|X_0 - C| \sin \varphi.$$

At $M_{u,v}$, the X -coordinate is

$$X_0 = \left(1 + \frac{c}{2}\right) u,$$

and

$$|X_0 - u| = |X_0 - (1 + c)u| = \frac{c|u|}{2}.$$

Thus, the distance from $M_{u,v}$ to either line $X = u$ or $X = (1 + c)u$ is

$$\frac{c|u|}{2} \sin \varphi.$$

Similarly, a line of the form $Y = C$ is parallel to the real axis. Hence, the Euclidean distance from $X_0 + Y_0\tau$ to the line $Y = C$ is

$$|Y_0 - C| |\tau| \sin \varphi.$$

At $M_{u,v}$, the Y -coordinate is

$$Y_0 = \left(1 + \frac{c}{2}\right) v,$$

so the distance from $M_{u,v}$ to either line $Y = v$ or $Y = (1 + c)v$ is

$$\frac{c|v||\tau|}{2} \sin \varphi.$$

Consequently,

$$d(M_{u,v}, \partial P_{u,v}) \geq \min \left\{ \frac{c|u|}{2} \sin \varphi, \frac{c|v||\tau|}{2} \sin \varphi \right\}.$$

Using the thickness assumptions,

$$d(M_{u,v}, \partial P_{u,v}) \geq \frac{c\eta \sin \varphi}{2}.$$

Set

$$\delta = \min \left\{ \frac{c}{4}, \frac{c\eta \sin \varphi}{4} \right\}.$$

Then

$$\delta < d(M_{u,v}, \partial A_{1,1+c})$$

and

$$\delta < d(M_{u,v}, \partial P_{u,v}).$$

Therefore,

$$B(M_{u,v}, \delta) \subset P_{u,v} \cap A_{1,1+c} = W_{u,v}.$$

□

Let

$$d_{\mathbb{T}}(\theta, \theta_0) = \min_{k \in \mathbb{Z}} |\theta - \theta_0 + 2\pi k|, \mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}.$$

Lemma 3.2. The ball $B(M_{u,v}, \delta)$ contains an annular sector

$$\{re^{i\theta} \mid r_1 < r < r_2, d_{\mathbb{T}}(\theta, \theta_0) < \varepsilon\},$$

where

$$\theta_0 = \arg(u + v\tau).$$

Proof. Set

$$M_{u,v} = \left(1 + \frac{c}{2}\right) e^{i\theta_0}.$$

Define

$$m = 1 + \frac{c}{2}, \quad r_1 = m - \frac{\delta}{2}, \quad r_2 = m + \frac{\delta}{2}, \quad \varepsilon = \frac{\sqrt{3}\delta}{2\sqrt{m(1+c)}}.$$

Choose any point $z = re^{i\theta}$, where $r_1 < r < r_2$ and $d_{\mathbb{T}}(\theta, \theta_0) < \varepsilon$. Observe that

$$|z - M_{u,v}|^2 = |re^{i\theta} - me^{i\theta_0}|^2 = (r - m)^2 + 2mr(1 - \cos(\theta - \theta_0)).$$

Since $r < 1 + c$ and $1 - \cos t \leq t^2/2$, we obtain

$$|z - M_{u,v}|^2 < \frac{\delta^2}{4} + m(1+c)\varepsilon^2 = \frac{\delta^2}{4} + \frac{3\delta^2}{4} = \delta^2.$$

Therefore, $z \in B(M_{u,v}, \delta)$, and hence

$$\{re^{i\theta} \mid r_1 < r < r_2, d_{\mathbb{T}}(\theta, \theta_0) < \varepsilon\} \subset W_{u,v}.$$

Thus, $W_{u,v}$ contains an annular sector of fixed radial and angular width, centred in the direction of α . $\square$

Lemma 3.3. There exist finitely many nonempty open angular intervals

$$I_0, \dots, I_{M-1} \subset \mathbb{R}/2\pi\mathbb{Z}$$

depending only on ε , such that, for every $\theta_0 \in \mathbb{R}/2\pi\mathbb{Z}$, there is an index j such that

$$I_j \subset \{\theta \mid d_{\mathbb{T}}(\theta, \theta_0) < \varepsilon\}.$$

Proof. Choose an integer M sufficiently large so that

$$\frac{2\pi}{M} < \varepsilon.$$

Define the open intervals

$$I_j = \left(\frac{2\pi j}{M}, \frac{2\pi(j+1)}{M} \right), \quad 0 \leq j < M.$$

Given any angle $\theta_0 \in \mathbb{R}/2\pi\mathbb{Z}$, there exists an index $j \in \{0, \dots, M-1\}$ such that

$$\frac{2\pi j}{M} \leq \theta_0 \leq \frac{2\pi(j+1)}{M}.$$

For any $\theta \in I_j$, the circular distance between θ and θ_0 is bounded by the length of the interval. Consequently,

$$d_{\mathbb{T}}(\theta, \theta_0) \leq \frac{2\pi}{M} < \varepsilon.$$

Thus, we obtain

$$I_j \subset \{\theta \mid d_{\mathbb{T}}(\theta, \theta_0) < \varepsilon\}.$$

□

Furthermore, this construction ensures that every I_j is an ordinary interval in $\mathbb{R}/2\pi\mathbb{Z}$, allowing Theorem 2.1 to apply directly.

Theorem 3.1. For every $c > 0$ and every $\eta > 0$, there exists $N_0(K, \tau, \eta, c)$ such that, if $\alpha \in \mathcal{O}_K$ is η -thick and

$$N(\alpha) \geq N_0(K, \tau, \eta, c),$$

then $B_\tau(\alpha, (1+c)\alpha)$ contains a prime element of $\mathcal{O}_K$.

Proof. Let $\theta_0 = \arg(\alpha)$. By Lemma 3.3, choose j such that

$$I_j \subset \{\theta : d_{\mathbb{T}}(\theta, \theta_0) < \varepsilon\}.$$

Apply the annular-sector result with $a = r_1^2$, $b = r_2^2$, and $x = N(\alpha) = R^2$. It gives a prime element ρ satisfying

$$r_1 R < |\rho| < r_2 R, \quad \arg \rho \in I_j.$$

Consequently ρ/R belongs to the sector in Lemma 3.2, which is contained in $W_{u,v}$. Hence $\rho \in \mathcal{B}_\tau(\alpha, (1+c)\alpha)$, and because $\rho \in \mathcal{O}_K$, it belongs to $B_\tau(\alpha, (1+c)\alpha)$, completing the proof. □

4 Further Directions

The thickness hypothesis ensures that the region contains an annular sector of uniform width. It would be useful to determine whether this hypothesis can be weakened.

Another problem is to obtain explicit bounds for $N_0(K, \tau, \eta, c)$, possibly under the generalized Riemann hypothesis.

References

- [1] Tchebichef, P. L., Mémoire sur les nombres premiers, *Journal de Mathématiques Pures et Appliquées*, 1852, 366–390.
- [2] Erdős, Paul, Beweis eines Satzes von Tschebyschef, *Acta Scientiarum Mathematicarum*, **5**(1932), 194–198.
- [3] Garcia, Stephan Ramon, Quotients of Gaussian Primes, *The American Mathematical Monthly*, **120**(2013), 851–853.
- [4] Hulse, Thomas A. and Murty, M. Ram, Bertrand's Postulate for Number Fields, *Colloquium Mathematicum*, **147**(2017), 165–180.
- [5] Klee, Steven, Loucks, Maiya, Meek, Samantha, Overcast, Levi, Stewart, A. J. and Tou, Erik R., Bertrand's Postulate over the Gaussian Integers, *The Pi Mu Epsilon Journal*, **14**(2018), 627–638.
- [6] Kubilius, Jonas, The Distribution of Gaussian Primes in Sectors and Contours, *Leningrad. Gos. Univ. Uc. Zap. Ser. Mat. Nauk*, **137**(1950), 40–52.
- [7] Rudnick, Zeév and Waxman, Ezra, Angles of Gaussian Primes, *Israel Journal of Mathematics*, **232**(2019), 159–199.
- [8] Sittertinger, Brian D., Quotients of Primes in an Algebraic Number Ring, *Notes on Number Theory and Discrete Mathematics*, **24**(2018), 55–62.
- [9] Ireland, Kenneth and Rosen, Michael, *A Classical Introduction to Modern Number Theory*, 2nd ed., Springer, 1990.